



OFFICIAL

Rhesus Negative and Clinically Significant Antibodies in Pregnant Women Procedure

1. Scope

Site	Operational Area	Applicable to
Armada Health Service	Obstetrics and Gynaecology Department	Nursing and Midwifery

2. Purpose

To determine the woman's ABO and Rh(D) blood group and to detect the presence of atypical red cell antibodies, the primary aim of which is to:

- Identify Rh(D) Negative women who may require the administration of prophylactic Rh(D) Immunoglobulin (RhD-Ig).
- To administer RhD-Ig to non-sensitised Rh(D) Negative women in a timely manner so the risk of maternal sensitisation to fetal Rh(D) Positive red blood cells is reduced.
- Identify women who are allo-immunised so they may undergo serological follow up and fetal assessment.
- Identify antibodies with potential to cause haemolytic disease of the fetus and newborn (HDFN).
- Provide selected, compatible blood for fetal and maternal transfusion when required.

3. Applicability

To guide staff on the procedure for administration of RhD-Ig (Anti-D) prophylaxis, given to prevent Rh(D) isoimmunisation in non-sensitised Rh(D) Negative women antenatally, postnatally and with pregnancy loss or sensitising events.

4. Guideline

Blood group incompatibility between a pregnant woman and her baby can lead to maternal antibody sensitisation and transfer significant antibodies across the placenta. In Australia 15-17% of pregnant women are RhD negative. It is estimated that only 37% of RhD negative women will carry a RhD negative fetus, meaning it is unnecessary to conduct routine antenatal administrations of RhD Ig to all women RhD negative.

RhD Ig is a human derived blood product with a limited supply source. The use of non-invasive prenatal testing (NIPT) with cell-free fetal DNA (cffDNA) to assess the fetal RhD status has been found to provide a safe and accurate assessment of the paternally inherited RhD blood type. Meaning RhD NIPT can be used to enable the targeted prophylaxis of RhD Ig to RhD negative women shown to be carrying a RhD positive fetus, rather than the routine antenatal administration to all RhD negative women.

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After birth cffDNA is rapidly cleared from the maternal circulation due to being derived from the placental trophoblast. The cffDNA can therefore be used as a pregnancy-specific marker as there is no risk of contamination from previous pregnancies. Using a maternal blood sample, which will contain both maternal and fetal cfDNA, current genomic methodologies enable a sufficient yield of cffDNA (known as the fetal fraction) to accurately predict the fetal RhD blood type.

Women who are identified as being RhD negative will be offered non-invasive prenatal testing to predict the RhD status of their babies which will allow for targeted antenatal RhD-Ig prophylaxis. A fetus who is predicted to be RhD positive or is unknown (e.g. not tested, result inconclusive, RHD NIPT declined) the RhD negative women will continue to be offered prophylactic antenatal RhD-Ig.

Women who do require antenatal prophylactic RhD-Ig can result in 78% reduction of alloimmunisation and administration at 28 and 34 weeks will result in reduction from 1% to 0.3% in alloimmunisation.

The administration of RhD-Ig as soon as possible and within 72 hours of a feto-maternal haemorrhage can remove fetal (RhD) positive cells from maternal circulation so that sensitisation does not occur. If not administered within 72 hours, a dose offered up to 9-10 days can still provide some protection.

4.1 Requirement to conduct Group and Screen

Women receiving antenatal care and intrapartum care at Armadale Health Service (AHS) will undergo blood group and antibody screening as outlined in the table below.

Gestation				
	Booking Appointment ➤ 20 weeks	28-30 weeks	34-36 weeks	On Admission
RhD Positive	Group and Screen (Review or order if not conducted in early pregnancy or no available results)	Group and Screen		Group and Screen (* See note below)
RhD Negative	Group and Screen (Review or order if not conducted in early pregnancy or no available results)	Current Group and Screen (The current group and screen must be collected within 2 weeks of a request for RhD Ig)	Current Group and Screen (If not done in this pregnancy the current group and screen must be collected within 2 weeks of a	Group and Screen (*see note below)

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	Offer NIPT		request for RhD Ig)	
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- * Current Group and Screen (G&S) is collected within two weeks of a request for RhD-Ig.
- A pre-birth Group and screen should be collected prior to all Induction of labours, if a woman is assessed as moderate to high risk as per the Risk Assessment – [Identifying Moderate to High Risk Women in Birth Suite Guideline](#) , or is booked for an Elective Caesarean Section (ELUSCS) or Non-Elective Caesarean Section (NELUSCS).

4.2 Pathology Form Documentation

The pathology form must include.

- The URMN sticker of the woman
- Any history of previous transfusion
- Gestation
- Previous history of antibodies, especially if reported outside of AHS
- Gravida and Parity
- Dates of any prophylactic RhD-Ig administered within the last 3 months.

4.3 Antenatal Care

- At the initial booking appointment establish the woman's ABO and RhD blood group and antibody screen. If a blood group and screen has not been performed the midwife is responsible for ordering these tests.
- The midwife will document the blood group, RhD and antibodies in the woman's handheld record and Stork database. Clearly mark in the woman's handheld notes that the woman is Rhesus Negative and send a copy of the group and screen pathology report for scanning into the Digital medical Records (DMR).
- If clinically significant antibodies are detected at the first antepartum visit, please follow steps below in Section 5.
- Women who are identified as being RhD negative it is recommended that these women are offered the RhD NIPT to determine the fetal RhD status. It is recommended to perform the NIPT with the woman's consent at 20-32 weeks gestation although preferably between 20-24 weeks gestation to ensure the appropriate administration of RhD-Ig at 28-32 weeks and 34-36 weeks gestation.
- If consented the midwife can request the RhD NIPT test through the pathology form.
- Women who are RhD negative and decline RhD NIPT or whom the RhD NIPT result is inconclusive will be managed as if the fetus is RhD positive and will receive prophylactic antenatal RhD-Ig.
- If the RhD NIPT is inconclusive it is not recommended to repeat the test.
- Women who sustain a potential sensitising event who are RhD negative, and it is predicted they have a RhD negative fetus through RHD NIPT testing, do not require RhD-Ig administration.
- Antenatal RhD-Ig prophylaxis will be provided to women who are RhD negative with no preformed anti D antibodies in whom RHD NIPT has predicted the fetus to be RhD positive will receive prophylactic RhD-Ig at 28-32 weeks and 34-36 weeks gestation and after any potentially sensitising events.

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- Women requiring prophylactic RhD-Ig during pregnancy will be provided on the initial antenatal booking visit the antenatal prophylaxis with Anti-D brochure.
- The midwife will complete the Obstetric Risk form AKMR 69 entering blood group and Anti D requirements.

4.4 Administration of Anti D procedure

RhD negative woman with Fetus predicted to be RhD positive or RHD test declined or inconclusive result			
28 weeks gestation	28-30 weeks	34-36 weeks gestation	Post Birth
Conduct routine blood group and antibody screen	Review blood group and antibody screen conducted at 28 weeks. • Administer first dose of prophylactic RhD-Ig 625 international units.	Review blood group and antibody screen conducted at 28 weeks. • Administer second dose of prophylactic RhD-Ig 625 international units.	• Collect Cord Blood to confirm neonate RhD positive. • Collect Maternal Fetal Haemorrhage • Administer RhD Ig (if neonate RhD positive)

- When RhD-Ig is administered the administrator must complete AKMR 176.20 Rh D Immunoglobulin (Anti D) Record including attaching the product label and barcode to the form and attaching the patients URMN sticker. This record will be scanned into the patients Digital Medical Record (DMR).
- RhD-Ig must be documented by a Health Practitioner on AKMR 170B Medication Chart prior to administration and signed by the practitioner administering the RhD-Ig.
- The administrator is responsible for completing the “Anti D register” when removing the RhD-Ig vial for administration to the woman from the vaccination fridge.
- In pregnancies less than 12 weeks gestation with a sensitising event RhD-Ig 250 International Units (IU) is administered e.g.
 - Chorionic Villus Sampling
 - Miscarriage
 - Ectopic pregnancy
 - Termination of pregnancy (medical or surgical)
- In twin pregnancies less than 12 weeks gestation with a sensitising event RhD-Ig 625 International Units is administered.
- Testing for anti-bodies at 28w prior to RhD-Ig administration is mandatory, but not required prior to 34week administration.
- If RhD-Ig is required, it must be given within 72 hours of birth or sensitizing event to the patient.

- Store RhD-Ig at 2-8°C. Do not freeze. RhD-Ig must only be removed from the fridge just prior to administration and brought to room temperature.
- Use immediately after opening the vial and discard any unused solution.
- RhD-Ig should be administered by deep intramuscular injection.
- For women with high BMI (e.g. BMI 30 or more), consideration should be given to factors which may impact on the adequacy of the injection, including the site of administration, access to underlying muscle and the length of the needle used.
- Two registered practitioners are responsible for the checking of the medication six rights of administration prior to administration and responsible to obtain consent before administering RhD-Ig.
- If the FMH test indicates a positive result, additional doses of RhD-Ig may be required after discussion with pathology and medical staff. RhD-Ig 625IU dose is enough to remove 6ml of fetal RhD positive red blood cells.

Practice Note: For every administration of RhD-Ig the administrator must complete the AKMR 176.20 Rh D Immunoglobulin (Anti D) Record. This will be scanned into DMR.

- A “Fetal Maternal Haemorrhage” (FMH) test is required for the following second and third trimester indications, optimally within an hour of the sensitising event – and RhD-Ig 625 IU given as required.
 - Amniocentesis
 - Fetal Death
 - Abdominal Trauma significant enough to cause FMH e.g. motor vehicle accident / domestic violence.
 - Antepartum haemorrhage (revealed or concealed)
 - External cephalic version (successful or unsuccessful)

There is no requirement to administer RhD-Ig injection if the woman is already sensitised. It is advisable to refer such woman to tertiary Fetal Maternal Medicine (MFM) for further management.

Women who decline RhD-Ig e.g. woman whose partner is also Rh negative, woman who decline blood products, woman indicates that she will not have any more babies, should be referred to specialist clinic for discussion and documentation of the discussion including risks and benefits.

4.5 Hospital admission procedure

- Access woman's DMR record, ICM pathology results and National Woman-Held Pregnancy Record MR70, and identify patients' blood group, antibody status, NIPT results and if RhD-Ig has been administered.
- A pre delivery Group and Hold sample should be collected on admission to Labour Ward if:
 - Atypical red cell antibodies are present

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- The woman's serology history is unknown
- Prophylactic RhD-Ig has been given
- There is increased risk of requiring a blood transfusion
- Prior to all Induction of labours, if a woman is assessed as moderate to high risk as per the Risk Assessment – [Identifying Moderate to High Risk Women in Birth Suite Guideline](#)
- Prior to all Elective Caesarean Section (ELUSCS)
- Prior to Non-Elective Caesarean Section (NELUSCS)

4.6 Post birth – collection of maternal and fetal blood, administration of Anti D

4.6.1 Collection of fetal blood post birth

A Blood Group and Direct Antiglobulin Test (DAT) should be collected on all babies born to a woman who is:

- Rh D Negative
- O Positive Blood Group (Refer to [AKG Neonatal Jaundice Management](#))
- Has known clinically significant antibodies
- Maternal Blood Group and Antibody Status unknown
- The primary accoucheur is responsible for the double clamping and collection of fetal cord blood, the blood is collected in two purple tubes and identified with the baby's UMRN labels and marked 'cord blood' with the date and time the specimen was taken and signed by the collector.
- The Fetal Cord blood is then sent with a pathology form requesting blood group and Direct Antiglobulin Test (DAT), with the baby's UMRN sticker, date, and time of collection with a matching collector's signature (as signed on the pathology bottles).

4.6.2 Collection of maternal blood post birth

- Maternal blood must be collected in a purple top pathology tube, no sooner than 15 minutes after birth to allow for the sensitising event and no later than 2 hours after the birth. Label with woman's UMRN labels and write the date and time the specimen was taken and signed by the collector.
- The maternal blood is sent to pathology with a completed pathology request form requesting Fetal Maternal Haemorrhage (FMH) and including the patients' blood group and confirming she has given birth. Include when RhD-Ig was administered in pregnancy on the pathology form. Include Maternal UMRN sticker, date and time of collection and the signature of the collector (must match signature on the pathology bottle).
- Report that bloods sent on both maternal ICM report and baby's ICM. Document in both maternal and neonatal DMR that bloods sent to pathology, and document on AKMR 176.20 Rh D Immunoglobulin (Anti D) Record on second page that bloods collected. Place AKMR 176.20 in the maternal blue patient folder.

4.6.3 Checking of pathology results post birth and administration of RhD-Ig

- Following the birth, handover should occur between the labour ward and the ward that the woman is Rh Negative and the maternal and cord bloods have been sent to pathology and require review when available.

- The postnatal unit coordinator is responsible for ensuring the pathology results are reviewed for both the mother and baby. If the baby is Rh positive, the coordinator is responsible for obtaining an order for RhD-Ig 625iU from the admitting practitioner written on AKMR 170B Medication Chart.
- The coordinator is responsible for confirming pathology results with the primary midwife/nurse.
- The primary midwife is responsible for obtaining the RhD-Ig, discussing the requirement for RhD-Ig with the woman, obtaining consent, administering the 625iU RhD-Ig and documenting the administration of RhD-Ig in the patients DMR, signing the medication chart and completing AKMR 176.20 Rh D Immunoglobulin (Anti D) Record. The primary midwife/nurse is responsible for documenting the administration of RhD-Ig in the register.

Baby's blood group must be confirmed as RhD positive prior to the administration of prophylactic RhD-Ig 625iU to a Rh D negative mother. A postnatal prescription is required prior to administration.

Practice Note: If a woman presents to Antenatal Assessment Unit with a sensitising event or labour ward with an unknown blood group take blood in a purple top EDTA tube and request an urgent blood group and antibody screen.

4.7 Contraindications to RhD-Ig administration:

- Rh positive individuals
- Rh negative individuals previously sensitised to Rh antigen (have pre-formed anti D antibodies)
- Severe thrombocytopenia or coagulation disorder enough to contra-indicate intramuscular injections
- Have isolated Immunoglobulin IgA deficiency, unless these women have had testing that shows they do not have circulating Anti-IgA antibodies

4.8 Precautions while administering RhD-Ig injection:

- If the woman's Body Mass Index (BMI) is ≥ 30 , product information recommends confirming clearance of fetal cells and the presence of RhD-Ig post administration. However, the consensus position statement recommends routine post administration testing may not be required unless there has been a large fetomaternal haemorrhage $>6\text{mls}$
- Prior systemic allergic reaction after administration of human immunoglobulin preparations
- Do not administer intravenously

5. Other significant antibodies causing HDFN:

Please refer to [KEMH Blood Group Management & Clinically Significant Antibodies: Rh D Negative & Rh D Positive Women](#) guideline for further management of antibodies outside of the scope of this guideline.

5.1 General principles of management of significant antibodies other than Rhesus negative:

If clinically significant antibodies are detected at the first antepartum visit, these antibodies will be identified, and a titration performed. Thereafter antibody investigation and titration should be repeated every 4 weeks until 28 weeks gestation and then every 2 weeks thereafter until delivery or as advised by TMU/ Australian Red Cross Blood Service (ARCBS) Red Cell Reference laboratory. The TMU will refer anti-D and anti-c antibodies to ARCBS for antibody quantitation.

All women with abnormal antibodies must have cross matched blood available on admission to Labour Ward as it takes time to cross match blood in such women.

A clinically significant rise in the antibody titre (or quantitation) will assist the clinician in determining when to initiate fetal monitoring, such as Doppler ultrasound and referral to tertiary Maternal Fetal Medicine (MFM) specialist.

The following are indications for referral to MFM Specialist:

- An antibody that may cause HDFN with the titre reaching or exceeding 1:32. The MFM Specialist and Haematologist/ Transfusion Medicine Unit (TMU) will advise on antibody significance
- Anti-K at any titre when the paternal phenotype is K positive
- All women who have had an infant previously affected by HDFN. These women should be referred to a specialist as soon as possible and preferably before 20 weeks gestation irrespective of antibody level. The partner's blood group and genotype should be obtained as early as possible in the pregnancy

6. Compliance, monitoring, and evaluation:

Audit of the use of Anti-D stickers for appropriateness of the timing and correct dosage of Anti-D injections.

7. Relevant standards

National Safety and Quality Health Service (NSQHS) Standards (second edition):

Standard 4: Action 4.3

Standard 7: Action 7.6

8. References

1. [Guideline for the prophylactic use of Rh D immunoglobulin in pregnancy care](#): National Blood Authority Australia. 2024
2. Module 5 Obstetrics and Maternity: patient Blood Management Guideline 2015 [National Blood Authority Australia](#)
3. RANZCOG College Statement: C-Obs 6 Guidelines for the use of Rh (D)

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- Immunoglobulin (Anti-D) in obstetrics in Australia: 2019
[HTTP://ranzcog.edu.au/wp-content/uploads/Anti-D-Guidelines.pdf](http://ranzcog.edu.au/wp-content/uploads/Anti-D-Guidelines.pdf)
4. [No. 343-Routine Non-invasive Prenatal Prediction of Fetal RHD Genotype in Canada: The Time is Here - ClinicalKey](#)
 5. [Non-invasive fetal RHD genotyping for RhD negative women stratified into RHD gene deletion or variant groups: comparative accuracy using two blood collection tube types - ClinicalKey](#) 2017
 6. Non-invasive prenatal testing (NIPT) for fetal Kell, Duffy and Rh blood group antigen predication in alloimmunised pregnant women: power of droplet digital PCR. [British Journal of Haematology | Wiley Online Library](#) 2020
 7. [Antenatal RHD NIPT for the Prediction of Fetal RhD Status](#). NMHS King Edward Memorial Hospital. May 2025
 11. [Blood Group and Antibody Screening in Pregnancy](#). NMHS King Edward Memorial Hospital. October 2023
 12. The Management of Women with Red Cell Antibodies during Pregnancy: Green Top Guideline No. 65 May 2014 https://www.rcog.org.uk/media/oykp1rtg/rbc_gtg65.pdf

9. Policy owner (relating to these procedures)

Dr Sangeeta Malla Bhat

Enquiries relating to this procedure may be directed to:

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10. Review

This procedure will be reviewed and evaluated as required to ensure relevance and currency. At a minimum it will be reviewed within one (1) year after first issue and at least every three (3) years thereafter.

Version	Effective from	Effective to	Amendment(s)
V4	24 10 2025	24 10 2025	Full review, New process of NIPT during pregnancy

11. Authorisation

This procedure has been authorised and issued by:

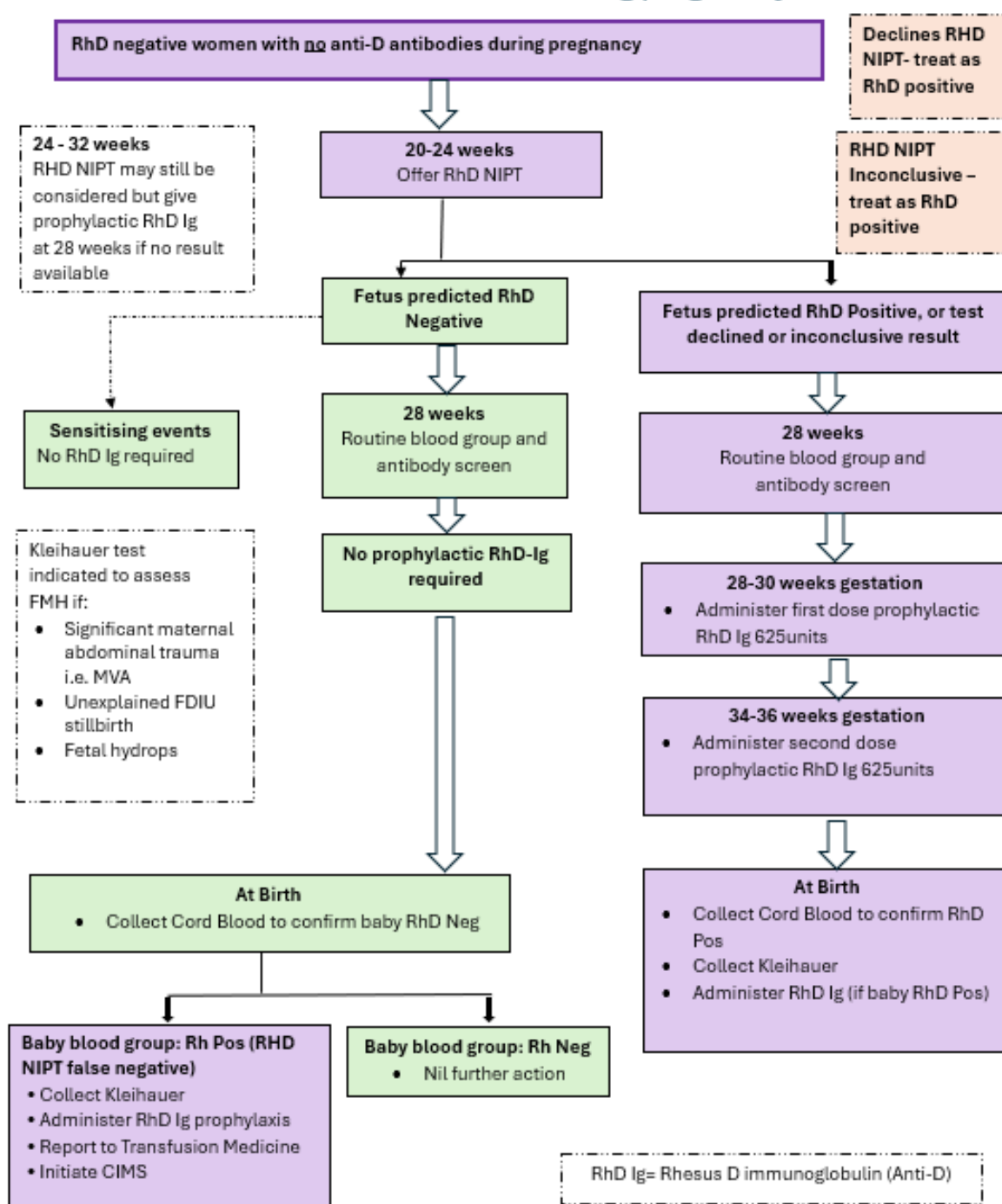
Authorised by	Dr Adrija Basu - Deputy Director Clinical Services
Authorisation date	23 10 2025
Published date	24 10 2025

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12. Appendices

12.1 Appendix 1: Management of RhD negative woman with no anti-D antibodies during pregnancy

Flowchart: RHD NIPT: Management of RhD negative women with no anti-D antibodies during pregnancy



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